

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is:

A **Denial of Service (DoS) attack**, specifically a **TCP SYN flood attack**, targeted the company's web server.

The logs show that:

The IP address **203.0.113.0** repeatedly sent a very large number of TCP **SYN** packets to the web server (192.0.2.1) without establishing normal connections. Legitimate users initially accessed the website successfully, but as the number of SYN requests increased, employees began receiving **504 Gateway Timeout** errors and **RST, ACK** responses instead of successful connections. Eventually, the server stopped responding to legitimate traffic and only recorded SYN requests from the attacker.

This event could be:

A **DoS SYN flood attack**. Because the malicious traffic originated from a single IP address rather than multiple systems, the attack is classified as a **DoS** attack rather than a **Distributed Denial of Service (DDoS)** attack.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol.

1. A client sends a **SYN** packet to request a connection.
2. The server responds with a **SYN-ACK** packet and reserves resources for the connection.
3. The client sends an **ACK** packet to complete the connection and begin communication.

In a SYN flood attack, the attacker continuously sends large numbers of SYN requests to

the web server. The server attempts to respond to each request and reserves memory and processing resources while waiting for the final ACK packet. As the number of SYN requests grows, the server's available resources become exhausted, preventing it from handling legitimate user requests.

The logs show repeated SYN packets from the attacker's IP address (203.0.113.0) being sent to port 443 on the web server. As the attack intensified, legitimate employee connections began failing, resulting in **RST, ACK** packets and **504 Gateway Timeout** errors. Eventually, the server became overwhelmed and could no longer process legitimate requests. Employees were unable to access the sales webpage, causing service disruption and reduced productivity.

Potential Consequences for the Organization

- Website outages and slow performance.
- Employees unable to access sales and vacation package information.
- Lost business opportunities and reduced customer service quality.
- Damage to the company's reputation and customer trust.
- Increased operational costs associated with responding to and mitigating the attack.

Recommendations to Prevent Future Attacks

- Enable **SYN flood protection** and rate limiting on firewalls and routers.
- Configure **SYN cookies** on the web server.
- Deploy an **Intrusion Detection/Prevention System (IDS/IPS)**.
- Use a **DDoS mitigation service** or cloud-based traffic filtering solution.
- Continuously monitor network traffic for unusual spikes in SYN requests.
- Implement firewall rules to automatically detect and block suspicious traffic patterns.

Attack Name: TCP SYN Flood (DoS Attack)

Impact on Network Performance: The attack overwhelmed server resources with excessive SYN requests, preventing legitimate users from establishing TCP connections and causing connection timeouts, failed requests, and website unavailability.